USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

1) Please fill out the Engineering Change Request Form on the following pages completely:
   a) Detail the names and contact details for each of the ECR contributors
   b) Update the ECR Title
   c) Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
2) For each section/table/figure to be updated:
   a) Detail the section number, starting page and figure/table number to be updated as appropriate.
   b) Detail existing text under "From Text"
   c) Detail changed text under "To Text"
3) Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step 1)b)
4) Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
   a) This ECR will then be reviewed by the Power Delivery Compliance Working Group.
   b) Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: Request Max Operating Current Update
Applied to: USB PD CTS Q3, 2026

Brief description of the functional changes proposed:
Update the CTS to align with the base specification definition of the maximum operating current in the Request Message. “Deprecated, transmitter Shall set this field equal to "Operating Current", receiver should ignore this field.”

Benefits as a result of the proposed changes:
Update CTS per base spec definition.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
Tester to update the test per the ECR.

An analysis of the Vendor Info File (VIF) implications:
None

Actual Change Requested

To Text:

COMMON.PROC.PD3.5 Tester Sent EPR_Mode (Enter)

Procedure Applicability: Unless otherwise specified, the Tester runs this procedure whenever the Tester is a Sink and tries to enter the EPR mode (0x01).

1. The Tester checks the UUT response as following:
   a. If the UUT responds with the Action field of the *EPR_Mode* Message set to the value 4(*Enter failed*), the Tester performs additional checks:
      i. If the Data field of the *EPR_Mode* Message set to value 5(*EPR Mode Capable bit not set in PDO*), the Tester checks VIF EPR_Supported_As_Src. If EPR_Supported_As_Src = Yes, the check fails, and the test stops here. Else, the check passes, and the test stops here. [COMMON.PROC.PD3.5#1]
      ii. If the Data field of the *EPR_Mode* Message set to value 4(*Source unable to Enter EPR Mode at this time*) (If it is the second instance of the response, the check fails, and the test stops here).
         1. If VIF Has_Invariant_PDOs set to Y, the Tester fails the check, and the test stops here [COMMON.PROC.PD3.5#2]
         2. Else, the Tester checks if the UUT can provide a higher power:
            a. The Tester sends the *Request* Message with the Capability Mismatch bit (B26) set to 1b, B19…10 (Operating Current / Power) is set to the Maximum Current / Power offered in the referenced Source PDO, and B9…0 (Maximum Operationg Current / Power(is set to the same value as B19…10. The Tester's higher power requirement is advertised in the Sink_Capabilities_Extended Message using VIF Src_PDO_Max_Power (see Table 16 Sink Capabilities Extended Message)

COMMON.CHECK.PD3.1 Check EPR_Request Message

Description: The Tester performs additional protocol checks to all *EPR_Request* Messages sent by the UUT.
Check Applicability: the *EPR_Request* Message
The Tester performs the following additional checks on all *EPR_Request* Messages: [COMMON.CHECK.PD3.1#1]

1. Field check for all types of Request Data Object
   a. B31…28 (Object Position) is not 000b, and the value is not greater than the number of PDOs in the last the *EPR_Source_Capabilities* Message
   b. B25 (USB Communications Capable), matches VIF field USB_Comms_Capable
   c. B24 (No USB Suspend), matches VIF field No_USB_Suspend_May_Be_Set
   d. Bit 23 (Unchunked Extended Messages Supported) matches VIF field Unchunked_Extended_Messages_Supported if the test is running in PD3 mode.
   e. Bits 22 (EPR Mode Capable) set to 1b.
   f. Bits 21 (Reserved) set to 0b.
2. Fixed and Variable Request Data Object
   a. B19…10 (Operating Current) does not exceed the current offered by the referenced PDO.
   b. B9…0 (Maximum Operating Current) is equal to B19…10 (Operating Current).
   i.

   c. B20 (Reserved) set to 0b.
3. Battery Request Data Object
   a. B19…10 (Operating Power) does not exceed the power offered by the referenced PDO.
   b. In PD3 mode, B9…0 (Maximum Operating Power) is equal to B19…10 (Operating Power).

COMMON.PROC.PD.18 Tester sent Request with Capability Mismatch for Full Capabilities Testing

Procedure Applicability: The Tester runs this procedure wherever COMMON.PROC.PD.18 is referred to in the test procedure.

1. The tester sends a *Request* Message as defined below:
   a. B30…B28 (Object Position) = Last Source Fixed PDO
   b. B27 (Reserved) = 0b
   c. B26 (Capability Mismatch) = 1b
   d. B25 (USB Communication Capable) = 0b
   e. B24 (No USB Suspend) = 1b
   f. B19…10 (Operating Current) = Maximum Current in Source PDO
   g. B9…0 (Maximum Operating Current) = 5A in PD2, Maximum Current in Source PDO in PD3.
2. If the *Wait* Message is received, the Tester re-sends the *Request* Message every *tSinkRequest* up to a maximum of *nBusyCount* times or until *Accept* is received.
3. If an *Accept* Message is not received, the check fails, and the test stops here.
4. If the first Preamble bit of *PS_RDY* Message is not received within *tSrcTransReq* from the last bit of the *GoodCRC* Message EOP corresponding to an *Accept* Message, the check fails, and the test stops here. [COMMON.PROC.PD.18#1]
5. The Tester responds to any *Get_Sink_Cap* Message with the *Sink_Capabilities* Message using the values found in the UUT VIF Source PDO values. It also responds to any *Get_Sink_Cap_Extended* Message using the values defined in **Error! Reference source not found.**.
6. If *Source Capabilities* are received with less than the *Sink Capabilities* (same as VIF Source Capabilities) and *tCapabilitiesMismatchResponse* has not elapsed, then the Tester goes back and repeats the *Request* Message in step 1.a with the Capability Mismatch bit =1.
7. Within *tCapabilitiesMismatchResponse,*
   a. If the *Source Capabilities* matching those in the VIF have been received by the Tester, it sends the *Request* Message defined in Step 1 with Capability Mismatch bit = 0. The check passes and proceeds to the next step. [COMMON.PROC.PD.18#2]

Check 1{

   b. If the *Source Capabilities* not matching in the VIF have been received by the Tester. [COMMON.PROC.PD.18#4]

      i. In the PD2 mode, proceed to the next step.
      ii. In the PD3 mode, the Tester sends the *Get_Source_Info* Message to the UUT and checks the response.
         1. If the PDP of the received *Source_Capabilities* is greater or equal to SIDO2 B8..0 (Port Guaranteed PDP), the test passes and proceeds to the next step.
         2. If the PDP of the received *Source_Capabilities* is greater or equal to the minimum of either Port Present PDP in the *Source_Info* Message or Sink Maximum PDP field in the *Sink_Capabilities_Extended* Message, the test passes and proeceeds to the next step, else the test fails and stops here.

Check 2 {

   c. If no Source Capabilities have been received by the Tester [COMMON.PROC.PD.18#5]

i. In the PD2 mode, proceed to the next step.
ii. In the PD3 mode,
    1. The Tester sends the *Get_Source_Info* Message to the UUT and checks the response.
    2. If SIDO1 B15…8 (Port Present PDP) equals to B7…0 (Port Reported PDP), the check passes and test proceeds to the next step.
    3. If any of the following conditions are met, the check fails, and the test stops here.
        a. SIDO1 B15…8(Port Present PDP) is not equal to B7…0 (Port Reported PDP), **or**
        b. the UUT does not reply with *Source Info,* **or**
        c. the UUT responds with a *Not Supported* Message.

} // Check 2

NOTE: It is expected that the internal Source state of the UUT is constant during this check, and it does not cause a change in Port Present PDP or Reported PDP .

8. If the *Wait* Message is received, the Tester re-sends the *Request* Message every *tSinkRequest* up to a maximum of *nBusyCount* times or until *Accept* is received.
9. If *Accept* Message is not received, the check fails, and the test stops here.
10. If the first Preamble bit of the *PS_RDY* Message is not received within *tSrcTransReq* from the last bit of the *GoodCRC* Message EOP corresponding to an *Accept* Message, the check fails, and the test stops here. [COMMON.PROC.PD.18#3]

COMMON.CHECK.PD.8 Check Request Message

Description: The Tester performs additional protocol checks to all *Request* Messages sent by the UUT.
Check Applicability: The *Request* Message
The Tester performs the following additional checks on all *Request* Messages: [COMMON.CHECK.PD.8#1]

4. Field check for all types of Request Data Object
    a. B31…28 (Object Position) is not 000b, and the value is not greater than the number of PDOs in the last *Source_Capabilities* Message
    b. B25 (USB Communications Capable), matches VIF field USB_Comms_Capable
    c. B24 (No USB Suspend), matches VIF field No_USB_Suspend_May_Be_Set
    d. PD3 mode:
        1. Bit 23 (Unchunked Extended Messages Supported) matches VIF field Unchunked_Extended_Messages_Supported
        2. Bit 22 (EPR Mode Capable),
            a. Always ‘0b’ if EPR_Supported_as_Snk=0
            b. Shall match the VIF field “EPR_Supported_As_Snk” in the last request prior entering the EPR mode.
        3. Bits 21 (Reserved) set to 0b .

    PD2 mode:
        Bits 23…20 (Reserved) set to 0000b.
5. Fixed and Variable Request Data Object
    a. B19…10 (Operating Current) does not exceed the current offered by the referenced PDO.
    b. B27 (Reserved) set 0b (formerly known as GiveBack). Also,
        i. In PD2 mode, if B9…0 (Maximum Operating Current) is greater than the current offered by the referenced PDO, then B26 (Capability Mismatch) is set to one.
        ii. In PD3 mode, B9…0 (Maximum Operating Current) is equal to B19…10 (Operating Current).
    c. B20 (Reserved) set to 0b.
6. Battery Request Data Object
    a. B19…10 (Operating Power) does not exceed the power offered by the referenced PDO.
    b. B27 (Reserved) set to 0b (formerly known as GiveBack). Also,

i. In PD2 mode, if B9…0 (Maximum Operating Power) is greater than the power offered by the referenced PDO, then B26 (Capability Mismatch) is set to one.
ii. In PD3 mode, B9…0 (Maximum Operating Power) is equal to B19…10 (Operating Power).

c. B20 (Reserved) set to 0b.

TEST.PD.PS.SRC.1 Multiple Request Load Test

Description: As a Sink only, the Tester verifies that the Source UUT responds correctly to various *Request* Messages.
Test Specific Tester Behavior: N/A
Test Conditions:

	Consumer Only, C/P	Provider Only, P/C, DRP
Rev2Src		✓
Rev3ChkdSrc		✓

Test Requirements: If Captive_Cable = YES, *vSrcNew* min, *vAvsNewmin*, or vPpsNew min limit shall be (*vSrcNew* or *vAvsNew* min) - (0.75*I/X) or (vPpsNew min) - (0.75*I/X), where I is the actual current being drawn, and X=3 if the sourced PDO contract is within the range (0, 3A], or X=5 if the sourced PDO contract is within the range (3A, 5A)

Test Procedures:

1. The Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.
2. The Tester repeats the following steps for each of the PDOs (specified in the VIF), starting with the first PDO:
   a. If the next Request will be for a PPS transition involving a current decrease (but no PDO change), the Tester reduces its current draw to the new level required, before sending the Request. (This avoids the current limit in the UUT shutting off V_{BUS}.) The Tester draws the requested Operating Current / Power with a current transition slew rate of 100mA per µs. The Tester monitors V_{BUS} voltage and the check fails if V_{BUS} voltage is not within the limits of *vPpsNew*. [TEST.PD.PS.SRC.1#9]
   NOTE: All measurements shall be within tolerances defined in the main specification and depicted in **Error! Reference source not found.**
   b. The Tester defines *Request* Message as following:
      i. B27 (Reserved) = 0b
      ii. B26 (Capability Mismatch) = 0b
      iii. B25 (USB Communication Capable) = 0b
      iv. B24 (No USB Suspend) = 1b
      v. B22 (EPR Capable) = 0b
      vi. For Fixed, Battery or Variable PDO:
         1. B19…10 (Operating Current / Power):
            a. For first Request: 0mA/0mW
            b. For subsequent Request: As defined in Step 2.f.i
         2. B9…0 (Maximum Operating Current / Power) = B19…10 (Operating Current / Power)
      vii. For PPS APDO:
         1. B20…9 (Output Voltage) = Maximum Voltage offered in Source APDO
         2. B6…0 (Operating Current):
            a. For first Request: 1A
            b. For subsequent Request: As defined in Step 2.f.ii
      viii. For SPR AVS APDO:

1. B20…9 (Output Voltage) – there are three iterations:
    a. 9V (minimum)
    b. 13.8V (an arbitrary value near the middle of the range)
    c. 20V if available, otherwise 15V (maximum)
2. B6…0 (Operating Current):
    a. For first Request: 0mA/0mW
    b. For subsequent Request: As defined in Step 2.f.i

The initial Request is for PDO #1 (5V Fixed) at 0mA. The complete list of Requests is defined in step 2.f.

TEST.PD.PS.SRC.4 Source Behavior with Capability Mismatch bit

Description: As a Sink only, the Tester verifies that the Source UUT responds correctly to various *Request* Messages with the Capability Mismatch bit set.
Test Specific Tester Behavior: N/A
Test Conditions:

	Consumer Only, C/P	Provider Only, P/C, DRP
Rev2Src		✓
Rev3ChkdSrc		✓

Test Requirements:

Test Procedures:

1. The Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.
2. The Tester repeats the following steps for each of the PDOs starting with the first PDO:
    a. The tester defines *Request* Message as follows:
        i. B27 (Reserved) = 0b
        ii. If Source Capabilities PDO Maximum Current < 5A or Maximum Power < 100W, B26 (Capability Mismatch) = 1b. If Source Capabilities PDO Maximum Current = 5A or Maximum Power = 100W, B26 (Capability Mismatch) = 0.
        iii. B25 (USB Communication Capable) = 0b
        iv. B24 (No USB Suspend) = 1b
        v. For Fixed, Battery or Variable PDO:
            B19…10 (Operating Current / Power) = Maximum Current / Power in Source PDO
            B9…0 (Maximum Operating Current / Power) = 5A /100W in PD2 mode; B19…10 (Operating Current / Power) in PD3 mode.
        vi. For AVS APDO:
            B20…9 (Output Voltage) = Maximum Voltage offered in the last Source Fixed PDO.
            B6…0 (Operating Current) = Maximum Current offered in Source AVS APDO.
            NOTE: In Source AVS APDO if B9…0 (Maximum Current) is set to 0, use B19…10 (Maximum Current) value.
        vii. For PPS:
            B20…9 (Output Voltage) = Maximum Voltage offered in Source APDO
            B6…0 (Operating Current) = Maximum current offered in Source APDO

TEST.PD.PS.SNK.2 Initial Sink PDO Transitions Post PR Swap

Description: The Tester verifies that the new Source UUT performs PDO transitions correctly.
Test Specific Tester Behavior:

- The Tester is a Dual-Role Power, initially a Source.

- The Tester sends *Sink_Capabilities* setting B27 (Unconstrained Power) to 0b COMMON.PROC.PD.3

Test Conditions:

	Consumer Only, C/P, DRP	Provider Only, P/C
Rev2Snk	✓	
Rev3ChkdSnk	✓	

Test Procedures:

1. The Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2
2. After this point the tester should *Reject* any *PR_Swap* request.
3. The Tester sends a *PR_Swap* Message.
4. The Tester checks the response: [TEST.PD.PS.SNK.2#1]
    a. The check fails if the UUT responds with a *Reject* Message or *Not_Supported*, and the VIF field Accepts_PR_Swap_As_Snk is "Yes".
    b. The test ends here if the UUT responds with a *Reject* Message or *Not_Supported*.
    c. If the UUT responds with a *Wait* Message, the Tester resends *PR_Swap* every *tPRSwapWait* + **Error! Reference source not found.** for ***Error! Reference source not found.*** times. Once the Tester has repeated for ***Error! Reference source not found.*** times, the check fails, and the test ends here.
    d. If the UUT responds with an *Accept* Message, the Tester checks that the VIF field PD_Port_Type is set to either "Consumer/Provider" or "DRP".
    e. If the UUT responds with an *Accept* Message, the Tester checks that the VIF field Accepts_PR_Swap_As_Snk is "Yes".
5. At *tSrcTransition* max (this delay is from the last bit of *GoodCRC* sent in response to the *Accept*), the Tester drives the V_{BUS} voltage to *vSafe0V* within *tSrcSwapStdby* max.
6. The Tester sends a *PS_RDY* at the deadline limit of *tPSSourceOff* min, the delay is from the time the last bit of the EOP of the *GoodCRC* corresponding to the *Accept* Message.
7. The Tester checks that the UUT sends *PS_RDY* only after the UUT has applied *vSafe5V*. [TEST.PD.PS.SNK.2#2]
8. The Tester checks that the *PS_RDY* from the UUT is within *tNewSrc* max, this delay is from the *GoodCRC* sent by the UUT in response to the *PS_RDY* sent by the Tester. [TEST.PD.PS.SNK.2#3]
9. Once the Tester receives *Source_Capabilities* Message, the Tester establishes a PD contract using common procedure COMMON.PROC.PD.11.
10. After this point the Tester shall *Reject* any *PR_Swap* request received.
11. The Tester repeats the following steps using the PDO as provided in the **PDO Sequence Table (**see Error! Reference source not found.**),** based on the Source PDOs specified in the VIF:
    a. The Tester defines the following *Request* Message:
        i. B27 (Reserved) = 0b
        ii. B26 (Capability Mismatch) = 0b
        iii. B25 (USB Communication Capable) = 0b
        iv. B24 (No USB Suspend) = 1b
        v. For Fixed PDO:
            B19…10 (Operating Current / Power) = 100mA / 0.5W
            B9…0 (Maximum Operating Current / Power) = B19…10 (Operating Current / Power).
        vi. For PPS PDO:
            B20…9 (Output Voltage) = Maximum Voltage offered in the last Fixed PDO
            B6...0 (Operating Current) = 1A
        vii. For SPR AVS APDO:
            B20…9 (Output Voltage) = Maximum Voltage offered in Source APDO
            B6...0 (Operating Current) = 1A

TEST.PD.PS.SNK.3 Multiple Request Load Test Post PR Swap

Description: The Tester verifies that the new Source UUT responds correctly to various *Request* Messages.
Test Specific Tester Behavior:

- The Tester is a Dual-Role Power, initially a Source.
- The Tester sends *Sink_Capabilities* setting B27 (Unconstrained Power) to 0b COMMON.PROC.PD.3

Test Conditions:

	C/P, DRP	Consumer Only, Provider Only, P/C
Rev2Snk	✓	
Rev3ChkdSnk	✓	

Test Requirements: If Captive_Cable = YES, *vSrcNew* min, *vAvsNew* min or *vPpsNew* min limit shall be (*vSrcNew* min or *vAvsNew* min) - (0.75*I/X) or (vPpsNew min) - (0.75*I/X), where I is the actual current being drawn, and X=3 if the sourced PDO contract is within the range (0, 3A], or X=5 if the sourced PDO contract is within the range (3A, 5A)
Test Procedures:

1. The Tester runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2
2. The Tester sends a *PR_Swap* Message.
3. The Tester checks the response: [TEST.PD.PS.SNK.3#1]
    a. The check fails if the UUT responds with a *Reject* Message or *Not_Supported*, and the VIF field Accepts_PR_Swap_As_Snk is "Yes".
    b. The test ends here if the UUT responds with a *Reject* Message or *Not_Supported*.
    c. If the UUT responds with a *Wait* Message, the Tester resends *PR_Swap* every *tPRSwapWait* + **Error! Reference source not found.** for ***Error! Reference source not found.*** times. Once the Tester has repeated for ***Error! Reference source not found.*** times, the check fails, and the test ends here.
    d. If the UUT responds with an *Accept* Message, the Tester checks that the VIF field PD_Port_Type is set to either "Consumer/Provider" or "DRP".
    e. If the UUT responds with an *Accept* Message, the Tester checks that the VIF field Accepts_PR_Swap_As_Snk is "Yes".
4. At *tSinkTransition* max (this delay is from the last bit of *GoodCRC* sent in response to the *Accept*), the Tester drives the V_{BUS} voltage to *vSafe0V* within *tSrcSwapStdby* max.
5. The Tester sends a *PS_RDY* at the deadline limit of *tPSSourceOff* min, the delay is from the time the last bit of the EOP of the *GoodCRC* corresponding to the *Accept* Message.
6. The Tester checks that the UUT sends *PS_RDY* only after the UUT has applied *vSafe5V*. [TEST.PD.PS.SNK.3#2]
7. The Tester checks that the *PS_RDY* from the UUT is within *tNewSrc* max, this delay is from the *GoodCRC* sent by the UUT in response to the *PS_RDY* sent by the Tester. [TEST.PD.PS.SNK.3#3]
8. Once the Tester receives *Source_Capabilities* Message, the Tester establishes a PD contract using common procedure COMMON.PROC.PD.11.
9. After this point the Tester shall reject any *PR_Swap* request received.
10. The Tester repeats the following steps for each of the PDOs (specified in the VIF), starting with the first PDO:
    a. If the next Request will be for a PPS transition involving a current decrease (but no PDO change), the Tester reduces its current draw to the new level required, before sending the Request. (This avoids the current limit in the UUT shutting off V_{BUS}.) The Tester draws the requested Operating Current / Power with a current transition slew rate of 100mA per µs. The Tester monitors V_{BUS}

voltage and the check fails if V_{BUS} voltage is not within the limits of *PpsNew*. [TEST.PD.PS.SNK.3#4]

b. The Tester defines *Request* Message as following:
   i. B27 (Reserved) = 0b
   ii. B26 (Capability Mismatch) = 0b
   iii. B25 (USB Communication Capable) = 0b
   iv. B24 (No USB Suspend) = 1b
   v. For Fixed, Battery or Variable PDO:
      1. B19…10 (Operating Current / Power):
         a. For first Request: 0mA/0mW
         b. For subsequent Requests: As defined in Step 10.f.i
      2. B9…0 (Maximum Operating Current / Power) = B19…10 (Operating Current / Power).
   vi. For PPS:
      1. B20…9 (Output Voltage) = Maximum Voltage offered in Source APDO
      2. B6…0 (Operating Current):
         a. For first Request: 1A
         b. For subsequent Requests: As defined in Step 10.f.ii.
   vii. For SPR AVS:
      1. B20…9 (Output Voltage) = Iterate through Min, Max and Mid Voltage offered in Source Fixed PDO
      2. B6…0 (Operating Current):
         a. For first Request: 0mA/0mW
         b. For subsequent Requests: As defined in Step 10.f.i.

The initial Request is for PDO #1 (5V Fixed) at 0mA. Further requests are defined in step 10f.

TEST.PD.PS.EPR.SRC3.1 Multiple EPR Request Load Test

Description: As a Sink only, the Tester verifies that the Source UUT responds correctly to various *EPR_Request* Messages.
Test Specific Tester Behavior:
Test Conditions:

This test is applicable if the VIF parameter EPR_Supported_*As_Src* is set to Yes.

	Consumer Only	Provider Only, P/C, C/P, DRP
Rev3ChkdSrc		✓

Test Requirements: If Captive_Cable = YES, *vSrcNew* min or *vAvsNew* min limit shall be (*vSrcNew* min) - (0.75*I/X) or (vAvsNew min) - (0.75*I/X), where I is the actual current being drawn, and X=3 if the sourced PDO contract is within the range (0, 3A], or X=5 if the sourced PDO contract is within the range (3A, 5A)

Test Procedures:

1. There are 2 possible bring-up procedures:
   a. The UUT has VIF field PD_Port_Type set to Consumer/Provider, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.7.
   b. The UUT has VIF field PD_Port_Type set to anything else, the Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.
2. The check fails if the "EPR mode capable" field is set to 0 in the *Source_Capabilities* Message sent by UUT during the bring up procedure and the test stops here. [TEST.PD.PS.EPR.SRC3.1#1]

3. The Tester sends an *EPR_Mode* (Enter) Message with the Action field set to 0x01(Enter) and Data field set to 140 (140W) and enters EPR mode using common procedure COMMON.PROC.PD3.5. [TEST.PD.PS.EPR.SRC3.1#2]
4. The Tester repeats the following steps for each of the EPR_PDOs only (in the EPR_*Source_Capabilities* Message), starting with the 8th PDO:
    a. The Tester sends an *EPR_Request* Message.
        i. B27 (Reserved) = 0b
        ii. B26 (Capability Mismatch) = 0b
        iii. B25 (USB Communication Capable) = 0b
        iv. B24 (No USB Suspend) = 1b
        v. B22 (EPR Mode Capable) = 1b
        vi. For Fixed PDO:
            1. B19…10 (Operating Current):
                a. For first *EPR_Request*: 0mA
                b. For subsequent Request: As defined in Step 4.f.i
            2. B9…0 (Maximum Operating Current) = B19…10 (Operating Current).
        vii. For EPR AVS APDO:
            1. B20…9 (Output Voltage) = Maximum Voltage offered in Source APDO
            2. B6…0 (Operating Current):
                a. For the first *EPR_Request*: 0mA
                b. For subsequent Request: As defined in Step 4.e.i

    The initial *EPR_Request* is for PDO #8 (28V Fixed) at 0mA. The complete list of *EPR_Request* is defined in step 4.e.

Spec References:

Request Message

6.4.2.1.1. Fixed and Variable RDO

Table 6.19. Fixed and Variable RDO

Bit(s)	Field Name	Static	Description
31...28	Object Position	No	See Section 6.4.2.1.5 .
27	Giveback	No	**Deprecated**, receiver **Shall** ignore this field.
26	Capability Missmatch	No	See Section 6.4.2.1.6 .
25	USB Communications Cable	Yes	See Section 6.4.2.1.7 .
24	No USB Suspend	Yes	See Section 6.4.2.1.8 .
23	Unchunked Extended Messages Supported	Yes	See Section 6.4.2.1.9 .
22	EPR Capable	No	See Section 6.4.2.1.10 .
21...20	Reserved	Yes	**Reserved**, receiver **Shall** ignore this field.
19...10	Operating Current	No	Operating current, in 10mA units Indicates the highest current the Sink will draw during the Explicit Contract. A new Request / EPR_Request Message, with an updated Operating Current value, **Shall** be issued whenever the Sink's power needs change. This value **Shall Not** exceed the value in the Maximum Current field of the Source_Capabilities Message. For EPR AVS, the Operating Current field **Shall Not** exceed the PDP ÷ Output voltage rounded down to the nearest 50 mA.
9...0	Maximum Operating Current	No	**Deprecated**, transmitter **Shall** set this field equal to "Operating Current", receiver should ignore this field.

6.4.2.1.2. Battery RDO

Table 6.20. Battery RDO

Bit(s)	Field Name	Static	Description
31...28	Object Position	No	See Section 6.4.2.1.5 .
27	Giveback	No	**Deprecated**, receiver **Shall** ignore this field.
26	Capability Missmatch	No	See Section 6.4.2.1.6 .
25	USB Communications Cable	Yes	See Section 6.4.2.1.7 .
24	No USB Suspend	Yes	See Section 6.4.2.1.8 .
23	Unchunked Extended Messages Supported	Yes	See Section 6.4.2.1.9 .
22	EPR Capable	No	See Section 6.4.2.1.10 .
21...20	Reserved	Yes	**Reserved**, receiver **Shall** ignore this field.
B19...10	Operating Power	No	Operating Power, in 250mW units Indicates the highest power the Sink will draw throughout the Explicit Contract.
B9...0	Maximum Operating Power	No	**Deprecated**, transmitter **Shall** set this field equal to "Operating Current", receiver should ignore this field.